

Customer Churn Dataset

Data Dictionary & Reference Guide

Telecom Industry | Customer Retention Analysis

Module 16 - MLOps Hands-on Lab | GUVI

Dataset Overview

This dataset simulates customer data from a telecommunications company. The primary business objective is to predict whether a customer will churn (leave the service) based on their usage patterns, contract details, and support interactions.

Customer churn is a critical problem in the telecom industry - acquiring new customers costs 5-25x more than retaining existing ones. By predicting which customers are likely to leave, companies can proactively offer incentives (discounts, upgrades, better plans) to retain them.

Quick Facts

Total Records:	1,000 customers
Total Columns:	11 (10 features + 1 target)
Target Variable:	churn (0 = Stayed, 1 = Left)
Churn Rate:	~35% (imbalanced dataset)
Industry:	Telecommunications
Use Case:	Binary Classification
ML Task:	Predict if a customer will churn

Column Descriptions

Below is a detailed description of every column in the dataset, including its data type, possible values, and what it represents in the business context.

1. customer_id	Type: String	Example: CUST_0001
A unique identifier assigned to each customer. Format: CUST_XXXX where XXXX is a zero-padded number from 0001 to 1000. <i>Business Insight: Used for tracking individual customers. Should be DROPPED before model training as it has no predictive value.</i> Possible Values: CUST_0001 to CUST_1000		
2. tenure	Type: Integer	Example: 52
The number of months the customer has been with the company. Ranges from 1 (brand new) to 72 (6 years). <i>Business Insight: Key predictor - new customers (low tenure) tend to churn more. Customers who stay beyond 12 months are much more likely to remain loyal.</i> Possible Values: 1 to 72 months		
3. monthly_charges	Type: Float	Example: 106.70
The amount (in USD) the customer pays each month for their telecom services. <i>Business Insight: Higher charges may correlate with churn, especially for customers who feel they are not getting value for money.</i> Possible Values: \$20.00 to \$120.00		
4. total_charges	Type: Float	Example: 5554.97
The total cumulative amount the customer has paid. Approximately tenure x monthly_charges with slight variation. <i>Business Insight: Derived metric that reflects both customer lifetime and spending level. Higher total charges usually indicate longer, more valuable customers.</i> Possible Values: Varies (tenure x monthly + noise)		
5. contract_type	Type: Categorical	Example: Two Year
The type of contract the customer is on. Determines how locked-in the customer is. <i>Business Insight: STRONG predictor of churn. Month-to-Month customers can leave anytime and have significantly higher churn rates.</i> Possible Values: Month-to-Month One Year Two Year		
6. internet_service	Type: Categorical	Example: Fiber Optic
The type of internet service the customer subscribes to. <i>Business Insight: Fiber Optic customers tend to have higher churn - often due to higher prices and service quality issues.</i> Possible Values: DSL Fiber Optic No		

7. online_security

Type: Categorical

Example: No

Whether the customer has an online security add-on service.

Business Insight: Customers with security add-ons tend to be more engaged and less likely to churn.

Possible Values: Yes | No

8. tech_support

Type: Categorical

Example: No

Whether the customer has a tech support add-on service.

Business Insight: Customers with tech support tend to churn LESS - they get issues resolved, leading to higher satisfaction.

Possible Values: Yes | No

9. payment_method

Type: Categorical

Example: Electronic Check

How the customer pays their monthly bill.

Business Insight: In real-world data, Electronic Check users tend to churn more (less commitment than auto-pay methods like Bank Transfer or Credit Card).

Possible Values: Electronic Check | Mailed Check | Bank Transfer | Credit Card

10. num_tickets

Type: Integer

Example: 3

Number of support tickets the customer has raised. Generated using a Poisson distribution (avg ~1.5).

Business Insight: Customers with many support tickets (>3) are frustrated and more likely to churn. This is a behavioural signal.

Possible Values: 0 to 9

11. churn

Type: Integer (Binary)

Example: 0

The TARGET variable. Indicates whether the customer has left the company.

Business Insight: This is what we are trying to PREDICT. 0 = Customer stayed (No Churn), 1 = Customer left (Churned).

Approximately 35% of customers in this dataset have churned.

Possible Values: 0 (No) | 1 (Yes)

Key Data Patterns & Churn Drivers

The dataset has been designed with realistic churn patterns that mirror what data scientists observe in actual telecom datasets. Understanding these patterns will help you interpret your model results better.

High Churn Risk Factors

- * Month-to-Month contract: +30% churn probability (no lock-in, easy to leave)
- * Fiber Optic internet: +15% churn probability (higher cost, service quality)
- * Low tenure (<12 months): +20% churn probability (new, not yet loyal)
- * High support tickets (>3): +15% churn probability (frustrated customer)

Low Churn Risk Factors

- * Tech support add-on: -10% churn probability (issues resolved, happier customer)
- * Long-term contract (1-2 Year): Only +5% base churn (committed customer)
- * High tenure (>12 months): No additional churn risk (loyal customer)

Real-World Insight

In actual telecom data, the churn formula is similar but more complex - it also includes factors like call drop rates, network coverage quality, competitor offers, and seasonality. Our simplified dataset captures the core patterns for learning purposes.

Feature Types Summary

Feature Type	Columns	Count
Identifier (drop before training)	customer_id	1
Numerical (continuous)	monthly_charges, total_charges	2
Numerical (discrete)	tenure, num_tickets	2
Categorical (multi-class)	contract_type, internet_service, payment	3
Categorical (binary)	online_security, tech_support	2
Target (binary)	churn	1

Preprocessing Steps Required

- * Drop customer_id - it is just an identifier, not a feature
- * Encode categorical columns using One-Hot Encoding (pd.get_dummies)
- * Scale numerical columns using StandardScaler for algorithms like Logistic Regression
- * Split data into 80% training / 20% testing with stratification on churn
- * Set random_state=42 everywhere for reproducibility

After Preprocessing

After one-hot encoding, the 10 features expand to ~16 columns (each category becomes its own binary column). The target variable "churn" remains as-is (already 0/1).